

VALERIO TANGARI

valerio.tangari@outlook.com ♦ +39 320 482 9589 ♦ [GitHub: valerioedu](#) ♦ [LinkedIn](#)

Curious and highly motivated first-year Physics Engineering student passionate about systems programming and low level software. Strong interests in Mathematics, Theoretical Physics and Motorsport.

EDUCATION

Politecnico di Milano

B.Sc. Physics Engineering

Milano

06/2024 - Present

Relevant Courses I'm taking:

- Fundamentals of Computer Sciences (Computer Programming)
- Algorithms and Data Structures
- Theoric Foundations of Computer Sciences
- Experimental Course: [Quantum Information Processing](#)

Liceo di Orbetello

Liceo Scientifico

Orbetello

09/2019 - 06/2024

TECHNICAL SKILLS

Programming Languages:

C, C++, C#, Go, Rust, Assembly, Python

Tools:

QEMU, Git, GitHub, Linux, Windows, TensorFlow, Qisqit

Concepts:

CI/CD, OS Development, Devops, Machine Learning

PROJECTS

x64OS

10/2024 - Present

A minimalistic open source AMD64 OS which aims to be Posix compliant.

Design and implementation of a 64-bit Unix-like operating system from scratch, focusing on low-level programming, memory management, and kernel architecture; running on QEMU. Key features include:

- Boot process: custom bootloader, GDT setup, paging, and long mode transition
- Memory: custom heap allocator (kmallocc, kfree, kreallocc) with aligned blocks and splitting/merging
- Kernel architecture: cooperative multitasking with idle loop
- Interrupts: PIC and PIT setup with IDT handling
- Filesystem: modular VFS layer with disk-based file system and POSIX-like syscalls (read, write, open, stat, sleep)
- Shell: command-line interface with support for cd, mkdir, touch, ls, etc.
- Drivers: VGA and Keyboard for I/O and IDE for disk

Skills: C, x64asm, QEMU, Make, Git

Some metrics:

- Single core CPU usage at just 20%, emulator included.
- Less than 100KiB binary size.
- 2MB paged Heap.
- Debugging enabled with kprint and kprintf

Chip-8 Emulator03/2025 - 03/2025

Created a complete Chip-8 Emulator in Rust using SDL, following [this guide](#).

Lossy Compression Evaluation — Winner of CERN-HSF GSoC 2025 Preselection03/2025 -03/2025

Analysis of lossy float compression (8/16-bit) on statistical metrics such as MSE and variance of main distributions in C++. Metrics are processed by a python script that plots: errors, binary size changes and evaluation comparisons. Won the CERN-HSF preselection for the 2025 Google Summer of Code proposal phase.

OPEN SOURCE AND HACKATHONS

utils coreutils03/2025 - 04/2025

Rust implementation of GNU coreutils.

- Built the edge case test for the arch command to handle unrecognized architectures.

MLH Cloud Hack Week03/2025 - 03/2025

Developed a chatbot using Python and the OpenAI API, leveraging snowflake and Microsoft Azure.

CERTIFICATIONS

- [GitHub Foundations](#)
- [Foundational C# with Microsoft](#)

LANGUAGES

Italian -Native
English -Fluent
Spanish -Limited Working Proficiency
German -Elementary Proficiency

